fy-m1: Teaching an Open Model to Chat in Everyday Malagasy

Version 0.1 — full human ratings in v0.2

by Fandresena Raharinaivo

October 2026

Abstract

Open language models that cover Malagasy tend to answer like an official letter: formal, stiff, and often half-translated from French or English. People in Madagascar do not text like that. I adapt an open instruction model, McGill-NLP/AfriqueQwen3.5-9B-50Langs-Instruct-v1 [McGill-NLP, 2026], to everyday Malagasy chat with QLoRA on a single GPU. The training mix combines 192 real family conversations that passed a strict automatic anonymisation gate, 1,284 synthetic chats that the base model itself wrote from few-shot real examples, short formal chats built from Malagasy Wikipedia, and identity chats that I wrote myself. I describe the privacy pipeline in full, because it is designed so that no person reads the private text, and I release the code, the evaluation prompts and every measured value through a script. The evaluation measures drift into full French or English, blind LLM-judge scores, a leak gate, an identity gate, and regressions on reading (Belebele) and translation (FLORES+). This is version 0.1: the base-model comparison, the data counts and the tokenizer trimming cost are measured; the fine-tuned model's scores print as "pending" until their result files exist, and the blind human comparison reports only early votes. No private text is released.

1 Introduction

Malagasy is the national language of Madagascar, but most Malagasy text online is formal: news, Wikipedia, religious and government writing. Models trained on that text learn that register. When a user writes a short, casual message, they get back a polished paragraph that no one would send to a friend.

Everyday Malagasy looks different. French words are mixed in freely, including French verbs used with Malagasy grammar ("lancer-na"). Spelling is phonetic and shortened ("zany" for "izany"). Replies are short, with humour and emoji. This is code-switching in the sense of Doğruöz et al. [2021], and for this project it is the target, not an error: a reply counts as drifting only if it is entirely French or English with no Malagasy in it.

My goal is modest and measurable: a model that is clearly better than its base model at everyday Malagasy chat, without losing much general Malagasy ability, trained on one GPU, served for free on a CPU, and released under a simple open licence. The constraint that shapes everything else is privacy: the only source of the target style is private family chat, and the people in it agreed to its use on the condition that nobody reads it. The contributions are:

1. a base-model comparison for Malagasy chat on a free 16 GB T4 GPU, with one Belebele protocol for every candidate and a licence rule stated up front (Section 4.1);
2. a privacy-first data pipeline in which no person reads the cleaned chats: anonymisation, a strict automatic gate that fails closed, a strict automatic gate on the synthetic chats, and a leak gate on the trained model (Sections 3 and 5.3);

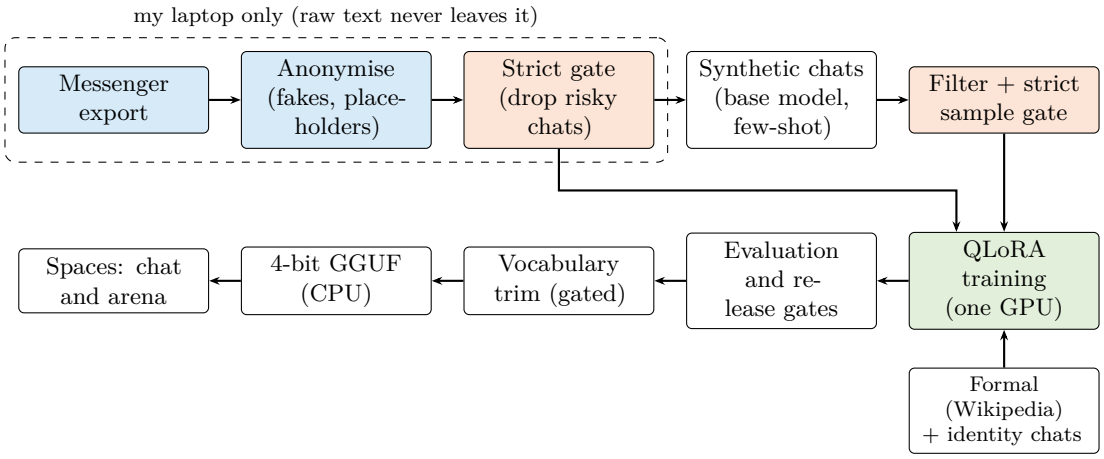


Figure 1: The fy-m1 pipeline. Blue steps run only on my laptop; orange steps are strict automatic gates that drop data instead of asking a person to read it. The real chats that pass the gate feed training directly and also serve as few-shot examples for the synthetic chats. The leak gate (Section 5.3) compares the trained model's answers with the real chats before release.

3. fy-m1 itself, with every number in this paper written by a script from a measured output file, so that a value not measured yet cannot appear.

Figure 1 shows the whole pipeline.

2 Related work

Malagasy in multilingual models. Malagasy is covered by large multilingual benchmarks such as Belebele [Bandarkar et al., 2024] and FLORES [Goyal et al., 2022, NLLB Team et al., 2022], now maintained as FLORES+ [Burchell et al., 2024, Open Language Data Initiative, 2024]. Recent open models list Malagasy among their languages, including the Gemma family [Gemma Team, 2025, Google, 2026] and AfriqueQwen, a Qwen-based model adapted to African languages [McGill-NLP, 2026]; multilingual instruction tuning at scale is studied by Üstün et al. [2024]. These benchmarks measure formal reading and translation; none measures casual chat, and I know of no public evaluation of informal, code-switched Malagasy.

Code-switching. Mixing languages within a conversation is the normal way many multilingual communities write, yet language technologies often treat it as noise [Doğruöz et al., 2021]. I take the opposite view for Malagasy–French and define drift only as a full switch away from Malagasy.

Parameter-efficient fine-tuning. LoRA [Hu et al., 2022] learns a low-rank update next to a frozen weight matrix; QLoRA [Dettmers et al., 2023] stores the frozen model in 4-bit NormalFloat (NF4), so a model of about ten billion parameters trains on one 16 GB GPU. I use Unsloth [Han et al., 2023], an implementation of the same method with lower memory use, and a cosine learning-rate schedule with linear warmup [Loshchilov and Hutter, 2017].

Synthetic data. Prompting a model to write its own training data is an established way to bootstrap instruction data [Wang et al., 2023]. Training a model repeatedly on its own outputs

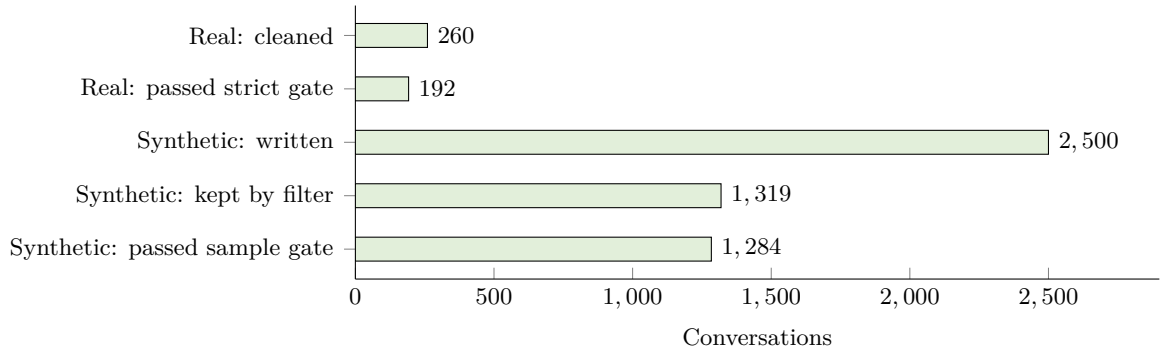


Figure 2: Data funnel (`paper/figdata/funnel.csv`, written by `paper/make_figures.py` from `results/`). Real: 260 cleaned conversations, 192 passed the strict gate. Synthetic: 2,500 written, 1,319 kept by the automatic filter, 1,284 passed the strict sample gate. Every step drops data on purpose; the formal and identity slices are not shown.

can narrow its output distribution [Shumailov et al., 2024]. My synthetic chats come from the base model itself (one round, few-shot prompted with real chats), so this risk applies; I discuss it in Section 7.

Privacy. Text anonymisation is hard to do completely, and masking named entities alone does not remove all identifying information [Lison et al., 2021]. Language models can also emit verbatim training text [Carlini et al., 2021]. I therefore combine anonymisation with gates that drop, rather than repair, anything uncertain, and with a verbatim-overlap check on the trained model.

Evaluation and compression. LLM judges are a cheap proxy for human preference [Zheng et al., 2023]; I use one blind and treat it as a signal, not a verdict. Language ID uses GlotLID [Kargaran et al., 2023], translation uses chrF++ [Popović, 2017]. Vocabulary trimming follows Ushio et al. [2023]; layer pruning [Gromov et al., 2025] is left for later.

3 Data

I describe the data but do not share it: part of it is private conversations. Only counts are published (`results/data_stats.json`, `results/data_gate.json`, `results/synth_stats.json`, `results/synth_gate.json`). Figure 2 shows how much data survives each step.

Real chats. The main source is one family's Messenger export (threads used: 1). Every sender's side is used as the assistant in turn. A local script splits the export into short multi-turn conversations at long gaps, merges back-to-back messages from the same sender, and replaces names, nicknames (also inside message text and handles), phone numbers, emails, links and addresses with fakes or placeholders (`docs/anonymisation.md`).

Strict automatic gate. No person reads the cleaned text; a strict automatic gate replaces the manual review I first planned. Figure 3 shows its decisions. At the word level, a word is kept only if a known-word list has it: common Malagasy words, Malagasy Wikipedia vocabulary [Wikimedia Foundation, 2023], Malagasy web text from HPLT 2.0 [Burchell et al., 2025], a short hand list of places and brands, and French and English words from `wordfreq` [Speer, 2022]. French first

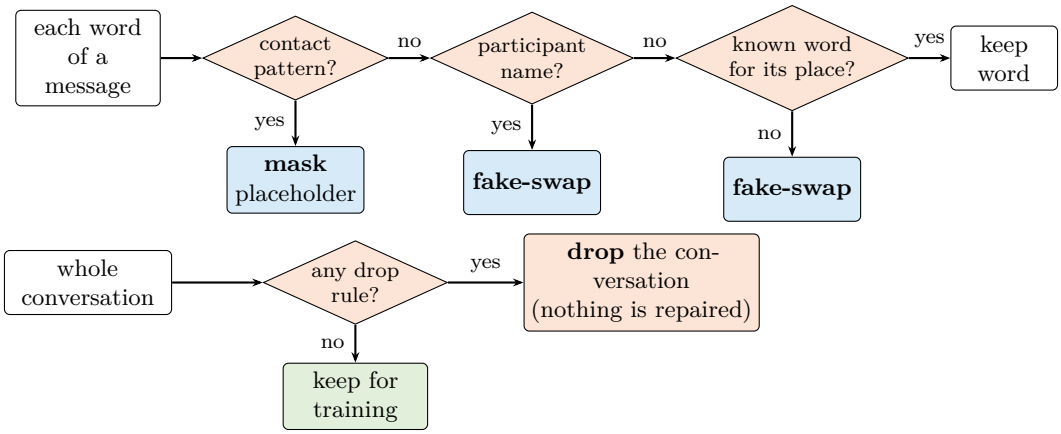


Figure 3: The privacy decisions, with no real example. Top: per word, the anonymiser masks contact patterns (phone, email, link, handle, address) with a placeholder and swaps participant names and nicknames for fakes; the strict gate then swaps any word that no known-word list clears for its position. Bottom: per conversation, the gate drops the whole conversation if any rule fires (long digit run, contact left, hidden character, more than 30% of words replaced, more than five fakes).

names from INSEE [INSEE, 2023] never count as known words. In a name position (mid-sentence or after a person marker such as *i* or *ry*) only the hand list can clear a word. Any other word is replaced with a fake name. At the conversation level the gate drops the whole conversation for a long digit run (a possible phone number), an email, link or handle, a hidden character, more than 30% of words replaced, or more than five fake names. Of 260 cleaned conversations it dropped 68 (a conversation can count under more than one reason): 0 for a long digit run, 4 for a contact, 15 for too many unknown words and 56 for too many fake names. The gate over-masks on purpose: 353 words were replaced in total, a median of 2 and at most 5 per kept conversation. 192 conversations (436 messages) passed.

Synthetic chats. Real chats are far too few on their own, so an open model that I run myself writes everyday chats, few-shot prompted with cleaned real chats. Only permissively licensed generators are allowed, because synthetic chats are training data and their licence reaches fy-m1: the code refuses any other. The generator was McGill-NLP/AfriqueQwen3.5-9B-50Langs-Instruct-v1 (licence cc-by-4.0), which is also the base model, so this step distils the style of the few-shot real examples back into the same model. An automatic filter kept 1,319 of 2,500 generated chats; the largest drop reasons were a word that no list clears in a name position (932), a length outside the allowed range (211), no Malagasy in a reply (23), and an 8-word overlap with the real chats (3). A strict automatic sample gate then checked all 1,319 kept chats for leftover real names, contacts, numbers, drift and broken turns, and dropped 35 (status: pass), leaving 1,284. No closed-API model output is in the training data.

Formal slice. **pending** short "tell me about" chats built from Malagasy Wikipedia paragraphs, so facts and standard spelling survive. The slice is kept small so it does not pull the style back to formal.

Identity chats. **pending** short exchanges, which I wrote in my own words, answer "who are you?" consistently (fy-m1, made by Fan's lab, built on an open model) and politely refuse harmful requests.

In total the model trained on **pending** examples (**pending** synthetic).

Consent and privacy of the data. Everyone in the conversations consented to their messages being used. The raw export never left my laptop; no person read the cleaned text, and only conversations that passed the strict gate were used. The cleaned data lives in a private dataset and is not released. The anonymiser has known limits, listed in `docs/anonymisation.md` and pinned as expected-failure tests: names of people outside the chat typed as plain text, a name that is also an everyday word at the start of a sentence, phone numbers written in words or split over two messages, and some addresses and place names. The strict gate and the leak gate (Section 5.3) are the backstops for these.

4 Method

4.1 Choosing the base model

I compared open instruction models that fit QLoRA training on a 16 GB T4 GPU, the free GPU on Kaggle and Colab. Each candidate was loaded in 4-bit and scored on: reading comprehension (Belebele, `plt_Latn`), both as plain-text 0-shot completion (one protocol for every candidate) and in the model's chat format; translation in both directions with English (FLORES+ devtest, chrF++); tokens per Malagasy word; a short QLoRA smoke run on a T4 (seconds per step); and a blind greeting gate, in which the judge of Section 5.2 scored 20 everyday greeting answers per model with model names hidden. A licence rule comes first: fy-m1 ships under a simple open licence, so the base must have one. Section 6.1 gives the measured values and the choice.

4.2 Fine-tuning with QLoRA

Adapters. For a frozen weight matrix $W_0 \in \mathbb{R}^{d\times k}$ of the base model, LoRA [Hu et al., 2022] learns $A \in \mathbb{R}^{r\times k}$ and $B \in \mathbb{R}^{d\times r}$ with rank $r \ll \min(d, k)$ and computes

$$h = W_0 x + \frac{\alpha}{r} BAx, \tag{1}$$

where α is a fixed scale. B starts at zero, so training starts from the base model exactly. QLoRA [Dettmers et al., 2023] stores W_0 in 4-bit NF4 and dequantises it on the fly; only A and B get gradients and optimiser state. A full update of W_0 trains dk weights, an adapter $r(d+k)$ (Figure 4). The adapter settings are rank $r = 16$ and $\alpha = 16$ (`results/lora_budget.json`, read from the training notebook's settings), on the attention and MLP projections of the language layers, using Unsloth [Han et al., 2023]. The base model is a hybrid: most of its layers use linear attention, interleaved with full-attention layers. Unsloth chooses which projections of each layer type get adapters, and the training log does not record the total trainable count, so I report the per-matrix size only.

Objective. Each training example is one conversation rendered in the base model's own chat template, with thinking turned off and a fixed Malagasy system prompt that names the model. Let $y_1, \ldots, y_T$ be its tokens and $m_t \in \{0, 1\}$ a mask that is 1 exactly on the tokens of assistant

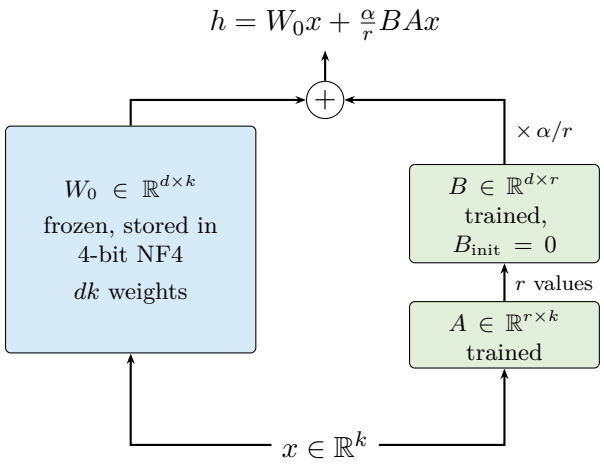


Figure 4: QLoRA on one weight matrix. Only A and B are trained, $r(d+k)$ weights. Example from the base model's configuration (`results/lora_budget.json`): an MLP up-projection has $d = 12,288$, $k = 4,096$, so $dk = 50,331,648$ weights; a rank-16 adapter trains 262,144 of them (0.52%). Peak GPU memory in the T4 smoke run was 9.8 GB (`results/model_selection.json`); the training run's peak memory was not logged.

	system prompt	user turn	assistant header	assistant reply	end of turn	user turn	assistant header	assistant reply	end of turn
tokens									
m_t	0	0	0	1	1	0	0	1	1

Figure 5: Assistant-only loss over the chat template (schematic, no real text). Green tokens have $m_t = 1$ and count in Equation 2; the rest are context only. Each assistant header includes the template's empty thinking block.

replies (each reply's end-of-turn token included; system prompt, user turns and assistant headers excluded). Training minimises the next-token cross-entropy on the masked tokens,

$$\mathcal{L}(A,B) = -\frac{1}{\sum_t m_t}\sum_{t=1}^{T} m_t \log p_{W_0,A,B}\left(y_t \mid y_{<t}\right), \tag{2}$$

averaged over the tokens of a batch (Figure 5). The template's generation prompt puts an empty thinking block after the assistant header; every assistant turn in training carries the same block, so the model learns the turn format it sees when it answers.

Optimisation. Training uses an 8-bit AdamW optimiser, a learning rate of **pending** with a **pending** schedule and **pending** linear warmup steps, sequence length **pending**, effective batch size **pending**, and **pending** epochs; these values are read back from the run's own log (`results/train_log.json`). A small held-out share of the conversations gives an evaluation loss at every checkpoint; training stops early when that loss rises at two evaluations in a row, and the checkpoint with the best evaluation loss is kept. Notebook sessions can disconnect, so checkpoints go to a private repository and a new session resumes from the latest one; a run refuses to resume if its settings or code changed. Training code reaches the GPU only as a synced copy of the clean main branch.

4.3 Making the model smaller: vocabulary trimming

Much of a multilingual model's vocabulary is for scripts and languages fy-m1 never uses. Following Ushio et al. [2023], I keep only the tokens seen in public Malagasy, French and English text (Wikipedia and HPLT 2.0, never the private chats) plus all special and byte-fallback tokens, and drop the matching embedding and output rows. The trim ships only if it passes three gates: held-out Malagasy text round-trips through the new tokenizer unchanged; greedy answers are unchanged for at least 95% of the test prompts; and Belebele stays within one point. Removing whole layers [Gromov et al., 2025] would save more but needs healing with extra training; I leave it for v0.2.

5 Experiments

5.1 Compute

Two kinds of GPU were used. Free T4 GPUs on Kaggle and Google Colab ran the base-model baselines, the synthetic-chat generation and the QLoRA smoke tests of Section 4.1. Training and evaluation of fy-m1 run on a paid Colab Pro A100 GPU; the training log records the GPU that was actually used (**pending**, `results/train_log.json`). The demo runs on a free CPU.

5.2 Test set and metrics

I generated answers from the base model and fy-m1 to **pending** fresh everyday prompts with the same system prompt and sampling settings. The prompts were written fresh, not taken from the chats, so the test set holds no private text and is public (`eval/prompts.jsonl`).

Drift. A reply counts as Malagasy if GlotLID [Kargaran et al., 2023] says so, or if at least a tenth of its words are common Malagasy words; mixed Malagasy–French passes. A reply that is fully French or English drifts. The drift rate is the share of replies that drift.

LLM judge. An LLM judge scored both answer sets blind, with labels hidden and order shuffled, on a published 1–5 rubric for naturalness and correctness [Zheng et al., 2023]. Methods disclosure: the judge is **pending**; rubric in `src/fym1/eval/judge_rubric.md`. The greeting gate of Section 4.1 used the same rubric with Claude Opus 5.5 (claude-opus-5-5), blind, run in Claude Code. The judge sees only the public prompts and the models' answers, never the private chats, and its outputs are never training data. Judge scores are a cheap signal, not a substitute for native speakers.

Regression checks. Belebele accuracy, FLORES+ chrF++ in both directions, and held-out chat perplexity should not get much worse after fine-tuning.

5.3 Release gates

Leak gate. Any fy-m1 answer that shares a run of 8 words with the real training chats blocks release, as a simple test for verbatim memorisation [Carlini et al., 2021]. The check is strict and automatic: only the matched words are looked at, never the chat around them, and any rejected run means fixing the data, filtering or retraining.

Identity check. A second strict, automatic gate covers the identity and safety prompts of the test set plus extra identity prompts in Malagasy, some mixed with French (`eval/identity_check.jsonl`). A reply passes when it is Malagasy (the drift test above) and names itself fy-m1 or Fan's lab as the prompt asks. Mentions of other models and companies are judged in two tiers. A *hard fail*, on any prompt, is an explicit false self-claim: another model's or company's name in a self-identification shape ("I am ...", "my name is ...", "made by ..." with the model as subject, in Malagasy, French or English), unless the shape is negated or only repeats the user's question. One hard fail fails the gate. An *ordinary miss*, only on prompts about the model itself, is any other unexcused mention of such a name; a denial or a repeated question excuses it. The split keeps the hard fail for clear lies about identity, while fuzzy wording only lowers the pass rate, so one odd phrasing cannot flip the release. No hard fail and at least **pending** of the **pending** replies passing are both required. The full word rules and their tests are in `src/fym1/eval/identity_check.py`. The gate checks identity only: whether the safety answers are good refusals is left to the judge rubric.

5.4 Blind human comparison

Native speakers rate pairs of answers in a public "arena": same prompt, two unlabelled answers in random order, pick the more natural one. Version 0.1 reports the early votes in at release, with a Wilson interval [Wilson, 1927]; the full panel's numbers come in v0.2.

5.5 Reproducibility

The test prompts, rubric, code and notebooks are public, and every number in this paper is a macro that `paper/make_numbers.py` writes from a file in `results/`; the plots read data that `paper/make_figures.py` writes from the same files. Version 0.1 releases only aggregate scores; the per-prompt answers of both models come in v0.2, after the human comparison closes, so raters cannot tell which model wrote which answer. The real training chats stay private, so training can be rerun only on other data. Training is a single run with one seed and no hyperparameter search.

6 Results

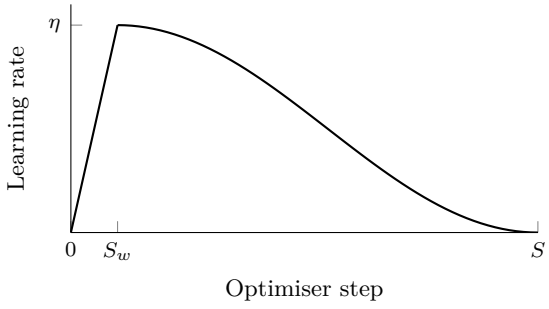
Every value in this section is written by `paper/make_numbers.py` from the `results/` file named next to it. A bold "pending" is a placeholder for a value that is not measured yet; it is filled in only when that file exists.

6.1 Base-model comparison

Table 1 shows the measured values (`results/model_selection.json`, `results/greeting_gate.json`). I chose McGill-NLP/AfriqueQwen3.5-9B-50Langs-Instruct-v1 (licence CC-BY-4.0). Among the models with measured translation scores it is best from English into Malagasy (chrF++ 49.7 versus 36.8 for Gemma 3 12B; Gemma 4 12B: **pending**), the direction closest to writing Malagasy. On the greeting gate it ties Gemma 4 12B (judge mean 2.90 versus 2.85 on 20 answers each; Malagasy replies 95.0% versus 90.0%), and chat-format Belebele is equal within noise (74.2% versus 75.0%). Its T4 smoke run took 83.3 seconds per step in 16-bit (Gemma 3: 122.4; Gemma 4: **pending**). The speeds are not a like-for-like comparison: on a T4, Unsloth runs some models in 32-bit, which is slower. I did not wait for Gemma 4's FLORES+ and smoke runs; it stays a comparison row, so the choice rests on a partial comparison.

Model	Licence	Belebele		chrF++		Tokens	Greeting
		text	chat	mg→en	en→mg	/word	judge (1–5)
McGill-NLP/AfriqueQwen-8B	CC-BY-4.0	**pending**	**pending**	**pending**	**pending**	2.95	**pending**
McGill-NLP/AfriqueQwen3.5-9B-50Langs-Instruct-v1	CC-BY-4.0	66.7%	74.2%	56.2	49.7	2.83	2.90
google/gemma-3-12b-it	Gemma Terms of Use	69.7%	75.1%	51.8	36.8	2.79	1.90
google/gemma-4-12B-it	Apache-2.0	**pending**	75.0%	**pending**	**pending**	2.79	2.85

Table 1: Base-model candidates, all loaded in 4-bit (`results/model_selection.json`, `results/greeting_gate.json`). Higher is better except tokens per word. "pending" means not measured. AfriqueQwen-8B is a base model: no chat score, not judged. Each greeting mean is over 20 answers, so small differences are not meaningful.



pending
loss curves: written from
results/train_log.json
when the training run ends

Figure 6: Left: learning rate, linear warmup over S_w steps to the peak η, then a half cosine to zero at the planned last step S [Loshchilov and Hutter, 2017]; recomputed from the run's settings in `results/train_log.json` once it exists (schematic until then). Right: training loss per logged step and held-out loss per evaluation, from the same log.

Gemma 3 12B [Gemma Team, 2025] is a measured baseline only: its terms of use would make fy-m1 a Gemma derivative, which the licence rule excludes. It also scored lowest on the greeting gate (judge mean 1.90). AfriqueQwen-8B was dropped early: it is a base model with no chat-tuned twin. A known risk of the chosen base is one reply in Hausa and some prompt echoes in its greeting answers; training on everyday chats targets exactly that.

6.2 Training

From `results/train_log.json`: **pending** steps, **pending** of training on **pending**. Final training loss was **pending**; best held-out loss was **pending**. Figure 6 shows the learning-rate schedule and the loss curves.

6.3 fy-m1 versus the base model

Table 2 compares the two models on the test set and the regression checks.

Measure	Base	fy-m1
Drift to full French/English (↓)	**pending**	**pending**
Naturalness, LLM judge, 1–5 (↑)	**pending**	**pending**
Correctness, LLM judge, 1–5 (↑)	**pending**	**pending**
Held-out chat perplexity (↓)	**pending**	**pending**
Belebele accuracy (↑)	**pending**	**pending**
FLORES+ chrF++ mg→en (↑)	**pending**	**pending**
FLORES+ chrF++ en→mg (↑)	**pending**	**pending**

Table 2: Base model versus fy-m1 on **pending** prompts (`results/eval/scores.json`; judge rows from `results/eval/judge.json`). Belebele and FLORES+ are regression checks: they should not fall much.

6.4 Release gates

Leak gate (`results/eval/scores.json`, `results/eval/leak_review.json`). **pending** of fy-m1 answers had a shared 8-word run; there were **pending** distinct matched runs, of which **pending** were judged generic phrases and **pending** were rejected.

Identity check (`results/eval/identity_check.json`). fy-m1 passed **pending** of **pending** replies (**pending**; threshold **pending**); hard fails: **pending**; gate **pending**.

6.5 Blind human comparison (early votes)

From `results/eval/arena.json`: **pending** votes from **pending** raters; fy-m1 win rate **pending** (95% Wilson interval **pending**–**pending**; Wilson, 1927). With so few raters this is a first signal, not a result.

6.6 Vocabulary trimming

The public keep set alone is measured (`results/shrink/trim_cost.json`): it shrinks the vocabulary from 248,077 to 74,040 tokens, and every held-out line decodes back unchanged. The cost is slightly longer token sequences: tokens per word go from 2.048 to 2.077 on the anonymised chats and from 2.269 to 2.302 on the test prompts. The trimmed model itself (`results/shrink.json`): gates **pending**; vocabulary **pending** → **pending** tokens; parameters **pending** → **pending**; download **pending** → **pending**; speed on the demo CPU **pending** → **pending** tokens/s.

7 Limitations

- **One family.** The real chats come from one family. The style reflects their everyday speech, not every region or dialect.
- **Mostly synthetic, from the base model itself.** There are 1,284 synthetic chats against 192 real ones, and the generator is the base model. Training a model on its own outputs can narrow what it says [Shumailov et al., 2024]; the few-shot real examples carry the new style, but the synthetic chats also carry the base model's habits and mistakes. The sample gate checks privacy and form, not quality, and is not a native speaker.
- **Over-masking.** The strict gate replaces rare words and texting spellings with fake names, which makes the real chats a little less natural.

- **Automatic evidence first.** The human comparison has only early votes from **pending** raters. Until the full panel is in, the main evidence is automatic.
- **The judge.** An LLM judge may not understand Malagasy slang well; its scores can be wrong in either direction, and I have not measured its agreement with native speakers. The judge is a closed model, so the judge scores cannot be reproduced exactly with open tools.
- **The drift metric is lenient.** English that contains a Malagasy name or a quoted Malagasy word can pass as Malagasy, so the drift rate can be a little low.
- **Small, single-run comparisons.** The greeting gate uses 20 answers per model, the base-model comparison is partial (Gemma 4's translation and speed are pending), and training is one run with one seed and untuned settings. No difference here comes with a significance test except the arena's interval.
- **Word rules for identity.** The identity gate matches word patterns; a false claim in an unusual phrasing can pass it.
- **A chat model, not a source of facts.** The model can be slow on a free CPU, get facts wrong, and invent things.

8 Ethics

Consent and privacy. Everyone in the chats agreed to the use of their messages. The pipeline is built so that no person reads the cleaned text, including me, and so that any doubt drops data instead of keeping it (Section 3). Neither the cleaned chats nor the raw answers are released in v0.1. Before release, the leak gate checks that the model does not repeat training text, and the public snapshot of the repository is checked automatically for names and copied chat text. The anonymisation is not perfect (Section 3 lists its known gaps); the gates reduce the risk but do not remove it.

Use of closed models. No closed-API model output is in the training data. The LLM judge (disclosed in Section 5.2) only scores answers to public prompts and never sees private text; its outputs are not training data.

Safety. The model was trained with a small set of polite refusals and gets a basic identity and safety check before release, but it is not a safety-tested product. A model that chats casually in a low-resource language is also harder for outside moderators to check, which is one more reason the model card states its limits plainly.

9 Future work

For v0.2:

- The full panel's blind human comparison, with confidence intervals, and the judge's agreement with it.
- Each model's answer to each test prompt.
- The layer-pruning experiment.

- Longer training, more than one seed, and more everyday data.

After that, two larger steps, in this order:

1. **A bigger curated corpus of everyday Malagasy text.** The main limit of fy-m1 is data: the real chats come from one family and most of the volume is synthetic. I plan to build a larger, curated and consented collection of everyday Malagasy writing, cleaned with the same strict privacy gates.

2. **A Malagasy speech model.** Much everyday Malagasy is spoken, not typed. With a stronger text model in place, I plan to work on a model that understands and speaks everyday Malagasy.

Acknowledgements

I thank everyone in the chats for agreeing to let me use their messages.

Licence. fy-m1's weights are released under CC-BY-4.0, the base model's licence (CC-BY-4.0), with attribution to McGill-NLP [2026]. The code is Apache-2.0. Gemma 3 was used only as a measured baseline; none of its outputs are in the training data. The formal chats are built from Malagasy Wikipedia text, which is CC-BY-SA 4.0. I credit it here. The weights stay CC-BY-4.0. Anyone who reuses Malagasy Wikipedia text must follow its ShareAlike terms.

Base model. AfriqueQwen3.5-9B-50Langs-Instruct-v1 by McGill-NLP [McGill-NLP, 2026], CC-BY-4.0. Gemma 4 12B [Google, 2026] (Apache-2.0) and Gemma 3 12B [Gemma Team, 2025] (Gemma Terms of Use) were comparison rows only.

Data. Data cleaning uses `wordfreq` by Robyn Speer [Speer, 2022] (word data CC-BY-SA 4.0; derived word lists keep that licence and attribution), word case counts from Malagasy Wikipedia [Wikimedia Foundation, 2023] (CC-BY-SA 4.0), HPLT 2.0 cleaned Malagasy web text [Burchell et al., 2025] (CC0-1.0 on its dataset card; I keep only a word list, never the text), and the INSEE first-name file [INSEE, 2023] (Licence Ouverte / Etalab 2.0). The formal slice uses Malagasy Wikipedia. Evaluation uses Belebele [Bandarkar et al., 2024] and FLORES+ [Open Language Data Initiative, 2024], both CC-BY-SA 4.0 on their dataset cards.

Tools. Language ID uses GlotLID [Kargaran et al., 2023], under the licence terms on its model card. Training uses Unsloth [Han et al., 2023] and Hugging Face Transformers, both Apache-2.0; the demo runs on llama.cpp (MIT). GPUs: free T4s from Kaggle and Google Colab, and a paid Colab Pro A100.

References

Lucas Bandarkar, Davis Liang, Benjamin Muller, Mikel Artetxe, Satya Narayan Shukla, Donald Husa, Naman Goyal, Abhinandan Krishnan, Luke Zettlemoyer, and Madian Khabsa. The belebele benchmark: a parallel reading comprehension dataset in 122 language variants. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages

749–775, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.44. URL https://aclanthology.org/2024.acl-long.44/.

Laurie Burchell, Jean Maillard, Antonios Anastasopoulos, Christian Federmann, Philipp Koehn, and Skyler Wang. Findings of the WMT 2024 shared task of the open language data initiative. In Barry Haddow, Tom Kocmi, Philipp Koehn, and Christof Monz, editors, *Proceedings of the Ninth Conference on Machine Translation*, pages 110–117, Miami, Florida, USA, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.wmt-1.4. URL https://aclanthology.org/2024.wmt-1.4/.

Laurie Burchell, Ona de Gibert, Nikolay Arefyev, Mikko Aulamo, Marta Bañón, Pinzhen Chen, Mariia Fedorova, Liane Guillou, Barry Haddow, Jan Hajič, Jindřich Helcl, Erik Henriksson, Mateusz Klimaszewski, Ville Komulainen, Andrey Kutuzov, Joona Kytöniemi, Veronika Laippala, Petter Mæhlum, Bhavitvya Malik, Farrokh Mehryary, Vladislav Mikhailov, Nikita Moghe, Amanda Myntti, Dayyán O'Brien, Stephan Oepen, Proyag Pal, Jousia Piha, Sampo Pyysalo, Gema Ramírez-Sánchez, David Samuel, Pavel Stepachev, Jörg Tiedemann, Dušan Variš, Tereza Vojtěchová, and Jaume Zaragoza-Bernabeu. An expanded massive multilingual dataset for high-performance language technologies (HPLT). In Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar, editors, *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 17452–17485, Vienna, Austria, July 2025. Association for Computational Linguistics. ISBN 979-8-89176-251-0. doi: 10.18653/v1/2025.acl-long.854. URL https://aclanthology.org/2025.acl-long.854/.

Nicholas Carlini, Florian Tramèr, Eric Wallace, Matthew Jagielski, Ariel Herbert-Voss, Katherine Lee, Adam Roberts, Tom Brown, Dawn Song, Úlfar Erlingsson, Alina Oprea, and Colin Raffel. Extracting training data from large language models. In *30th USENIX Security Symposium (USENIX Security 21)*, pages 2633–2650, 2021. URL https://www.usenix.org/conference/usenixsecurity21/presentation/carlini-extracting.

Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. QLoRA: Efficient finetuning of quantized LLMs. In *Advances in Neural Information Processing Systems*, volume 36, 2023. URL https://arxiv.org/abs/2305.14314.

A. Seza Doğruöz, Sunayana Sitaram, Barbara E. Bullock, and Almeida Jacqueline Toribio. A survey of code-switching: Linguistic and social perspectives for language technologies. In Chengqing Zong, Fei Xia, Wenjie Li, and Roberto Navigli, editors, *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 1654–1666, Online, August 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.acl-long.131. URL https://aclanthology.org/2021.acl-long.131/.

Gemma Team. Gemma 3 Technical Report. *arXiv preprint arXiv:2503.19786*, 2025.

Google. Gemma 4 12B Instruct. Hugging Face model card, https://huggingface.co/google/gemma-4-12B-it, 2026. Licence Apache-2.0.

Naman Goyal, Cynthia Gao, Vishrav Chaudhary, Peng-Jen Chen, Guillaume Wenzek, Da Ju, Sanjana Krishnan, Marc'Aurelio Ranzato, Francisco Guzmán, and Angela Fan. The Flores-101 evaluation benchmark for low-resource and multilingual machine translation. *Transactions of the Association for Computational Linguistics*, 10:522–538, 2022. doi: 10.1162/tacl_a_00474. URL https://aclanthology.org/2022.tacl-1.30/.

Andrey Gromov, Kushal Tirumala, Hassan Shapourian, Paolo Glorioso, and Daniel A. Roberts. The unreasonable ineffectiveness of the deeper layers. In *International Conference on Learning Representations*, 2025. URL https://arxiv.org/abs/2403.17887.

Daniel Han, Michael Han, and Unsloth team. Unsloth. https://github.com/unslothai/unsloth, 2023.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022. URL https://arxiv.org/abs/2106.09685.

INSEE. Fichier des prénoms. https://www.insee.fr/fr/statistiques/7633685, 2023. Licence Ouverte / Etalab 2.0.

Amir Hossein Kargaran, Ayyoob Imani, François Yvon, and Hinrich Schuetze. GlotLID: Language identification for low-resource languages. In Houda Bouamor, Juan Pino, and Kalika Bali, editors, *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 6155–6218, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.findings-emnlp.410. URL https://aclanthology.org/2023.findings-emnlp.410/.

Pierre Lison, Ildikó Pilán, David Sanchez, Montserrat Batet, and Lilja Øvrelid. Anonymisation models for text data: State of the art, challenges and future directions. In Chengqing Zong, Fei Xia, Wenjie Li, and Roberto Navigli, editors, *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 4188–4203, Online, August 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.acl-long.323. URL https://aclanthology.org/2021.acl-long.323/.

Ilya Loshchilov and Frank Hutter. SGDR: Stochastic gradient descent with warm restarts. In *International Conference on Learning Representations*, 2017. URL https://arxiv.org/abs/1608.03983.

McGill-NLP. AfriqueQwen3.5-9B-50Langs-Instruct-v1. Hugging Face model card, https://huggingface.co/McGill-NLP/AfriqueQwen3.5-9B-50Langs-Instruct-v1, 2026. Licence CC-BY-4.0.

NLLB Team, Marta R. Costa-jussà, James Cross, Onur Çelebi, Maha Elbayad, et al. No language left behind: Scaling human-centered machine translation. *arXiv preprint arXiv:2207.04672*, 2022. URL https://arxiv.org/abs/2207.04672.

Open Language Data Initiative. FLORES+. Hugging Face dataset, https://huggingface.co/datasets/openlanguagedata/flores_plus, 2024.

Maja Popović. chrF++: words helping character n-grams. In Ondřej Bojar, Christian Buck, Rajen Chatterjee, Christian Federmann, Yvette Graham, Barry Haddow, Matthias Huck, Antonio Jimeno Yepes, Philipp Koehn, and Julia Kreutzer, editors, *Proceedings of the Second Conference on Machine Translation*, pages 612–618, Copenhagen, Denmark, September 2017. Association for Computational Linguistics. doi: 10.18653/v1/W17-4770. URL https://aclanthology.org/W17-4770/.

Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Nicolas Papernot, Ross Anderson, and Yarin Gal. AI models collapse when trained on recursively generated data. *Nature*, 631:755–759, 2024. doi: 10.1038/s41586-024-07566-y.

Robyn Speer. wordfreq. https://github.com/rspeer/wordfreq, 2022. Code Apache-2.0, word data CC-BY-SA 4.0.

Asahi Ushio, Yi Zhou, and Jose Camacho-Collados. Efficient multilingual language model compression through vocabulary trimming. In Houda Bouamor, Juan Pino, and Kalika Bali, editors, *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 14725–14739, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.findings-emnlp.981. URL https://aclanthology.org/2023.findings-emnlp.981/.

Ahmet Üstün, Viraat Aryabumi, Zheng Yong, Wei-Yin Ko, Daniel D'souza, Gbemileke Onilude, Neel Bhandari, Shivalika Singh, Hui-Lee Ooi, Amr Kayid, Freddie Vargus, Phil Blunsom, Shayne Longpre, Niklas Muennighoff, Marzieh Fadaee, Julia Kreutzer, and Sara Hooker. Aya model: An instruction finetuned open-access multilingual language model. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15894–15939, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.845. URL https://aclanthology.org/2024.acl-long.845/.

Yizhong Wang, Yeganeh Kordi, Swaroop Mishra, Alisa Liu, Noah A. Smith, Daniel Khashabi, and Hannaneh Hajishirzi. Self-instruct: Aligning language models with self-generated instructions. In Anna Rogers, Jordan Boyd-Graber, and Naoaki Okazaki, editors, *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 13484–13508, Toronto, Canada, July 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.acl-long.754. URL https://aclanthology.org/2023.acl-long.754/.

Wikimedia Foundation. Wikipedia dataset (wikimedia/wikipedia). Hugging Face dataset, https://huggingface.co/datasets/wikimedia/wikipedia, 2023. Text CC-BY-SA 4.0.

Edwin B. Wilson. Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158):209–212, 1927.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Advances in Neural Information Processing Systems (Datasets and Benchmarks Track)*, volume 36, 2023. URL https://arxiv.org/abs/2306.05685.